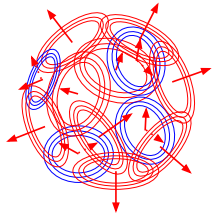
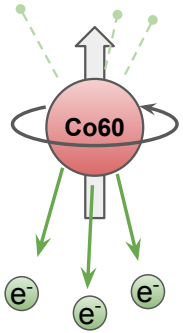
R-space Theory 1.0

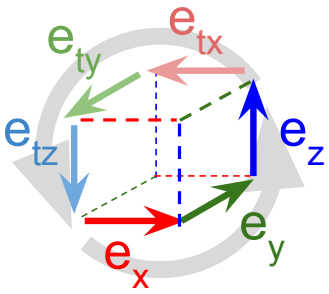
- Geometric Integration of Relativity and Quantum Mechanics -



Chap.10: Parity Symmetry Violation

What are Neutrinos? Part 2





1. Parity Symmetry Violation in the Weak Force

The weak force, which is involved in processes such as neutrino generation in β-decay, violates parity symmetry.

In physics, parity refers to symmetry under spatial inversion (mirror reflection); parity is conserved in electromagnetism, the strong force, and gravity.

The experimental verification of parity symmetry violation in the weak force was conducted in 1957 by Chien-Shiung Wu and colleagues, in what is known as the Wu experiment.

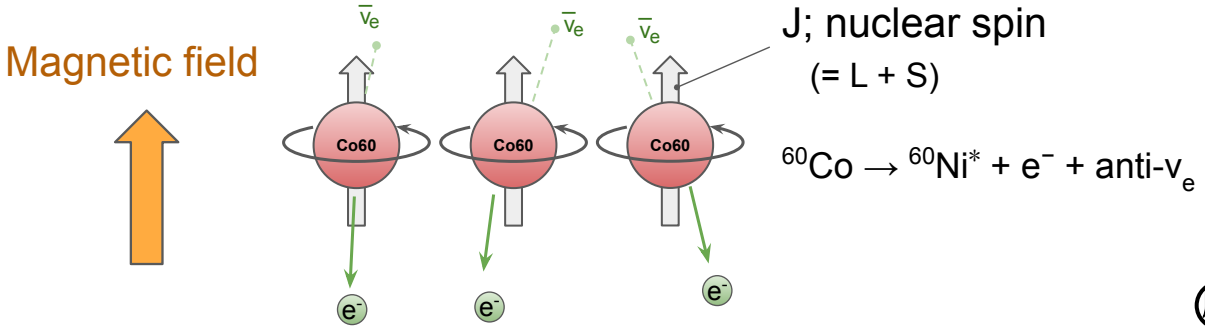
In the Wu experiment, Cobalt-60 was placed at ultra-low temperatures under a strong magnetic field to align nuclear spins in a single direction, and the directional distribution of electrons emitted via β-decay was measured.

1. Parity Symmetry Violation in the Weak Force

If parity were conserved in the weak force, electrons should have been emitted equally in both directions relative to the nuclear spin.

However, the experimental results revealed that electrons were emitted preferentially in one direction—opposite to the nuclear spin direction.

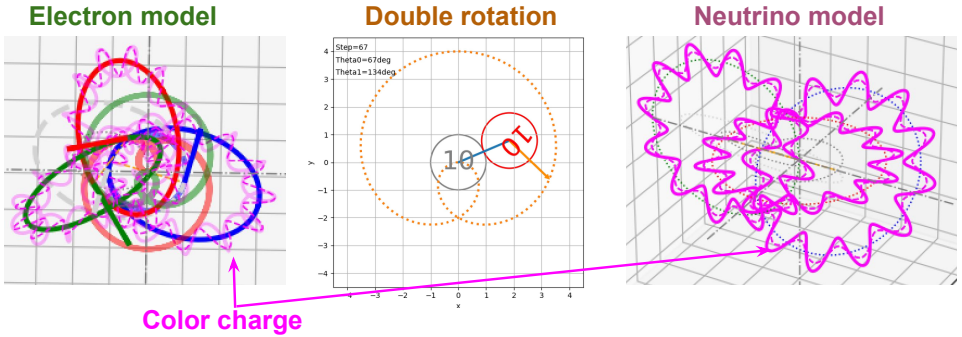
In present quantum mechanics, parity symmetry violation in the weak force is explained as arising because the weak force acts exclusively on the left-chiral component of particles.



1. Parity Symmetry Violation in the Weak Force

However, the assertion that the weak force acts only on the left-chiral component is an empirical finding derived from experimental results; why this is the case remains unknown.

In R-space Theory, the mechanism behind parity symmetry violation in the weak force is explained through the structure of massive particles—namely, "double rotation in a Hopf-link structure" and "the relationship between neutrinos and color charge".



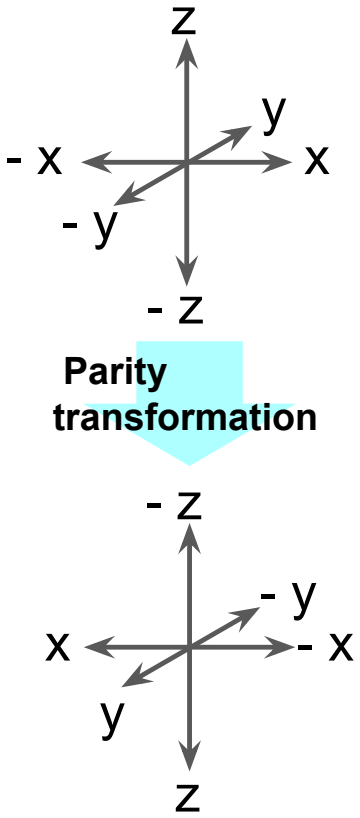
2. What is Parity Symmetry?

Parity symmetry violation in the weak force involves complex interactions between relativity, chirality, and helicity.

First, the mechanism of parity symmetry violation in current quantum mechanics is explained.

Parity symmetry refers to symmetry under spatial inversion (mirror reflection), which differs slightly from the left-right reversal seen in everyday mirror reflections.

A parity transformation replaces coordinates (x, y, z) with (-x, -y, -z).



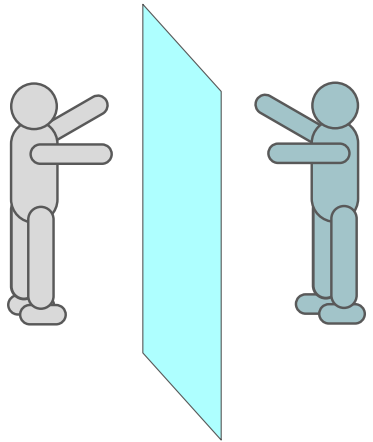
2. What is Parity Symmetry?

What appears as left-right reversed in an actual mirror is a artifact of human perception; as a coordinate transformation, it is a "front-back inversion (sign reversal in the depth direction)", whereas top-bottom and left-right coordinates undergo no sign reversal.

Conservation of parity means that physical laws remain invariant under a parity transformation.

For example, consider a ball thrown from left to top-right.

The ball moves parabolically according to the equation of motion “F = ma”.



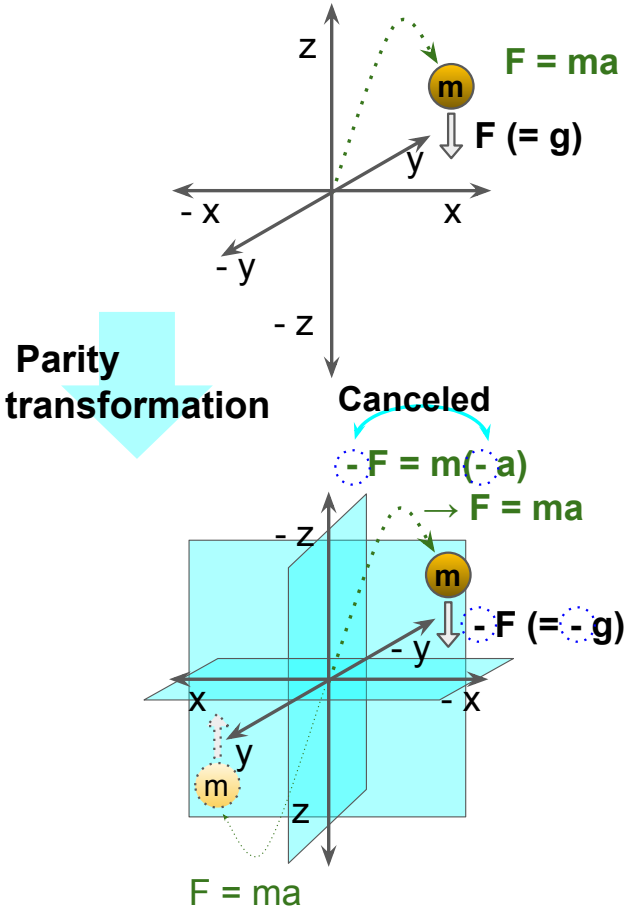
2. What is Parity Symmetry?

Under parity transformation, the coordinates change as “x to -x”, “y to -y”, “z to -z”.

Because coordinate directions change, the signs of both the applied force (F = gravity; g) and acceleration flip, yielding “-F = m(-a)”.

The negative signs on both sides cancel out, leaving the equation expressing the physical law identical to its pre-transformation form (same equation = physical laws unchanged).

Thus, parity is conserved in Newtonian mechanics.



2. What is Parity Symmetry?

Furthermore, experimental evidence confirms parity conservation in electromagnetism (atomic spectral lines, light scattering, electron scattering, electromagnetic transitions in nuclei) and in the strong force (meson scattering, hadron decay, excited nuclear state transitions).

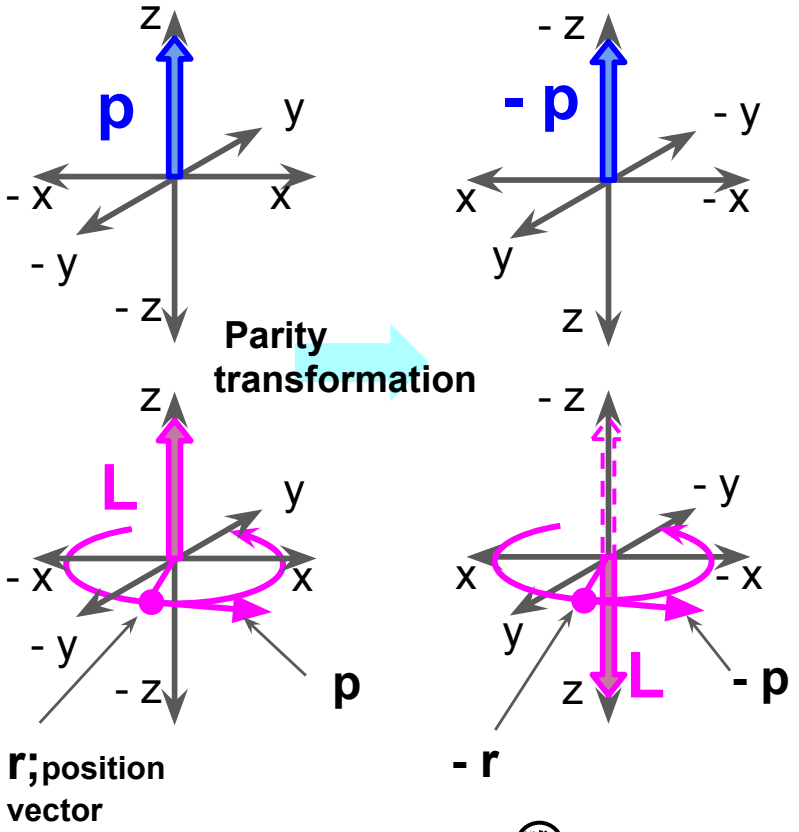
Consequently, parity was long believed to be universally conserved in physics.

In the Wu experiment, the fundamental difference in behavior between momentum and angular momentum under parity transformation was utilized to test parity violation in the weak force.

2. What is Parity Symmetry?

Momentum (p) flips sign under parity transformation (p to -p). However, angular momentum (L = p × r) involves the position vector “r”, which also flips sign, so the sign reversals cancel out (L = -p × -r ⟶ L = p × r).

Vectors possessing this property are called pseudovectors (axial vectors), and under parity transformation, their directional arrows invert as shown in the diagram.

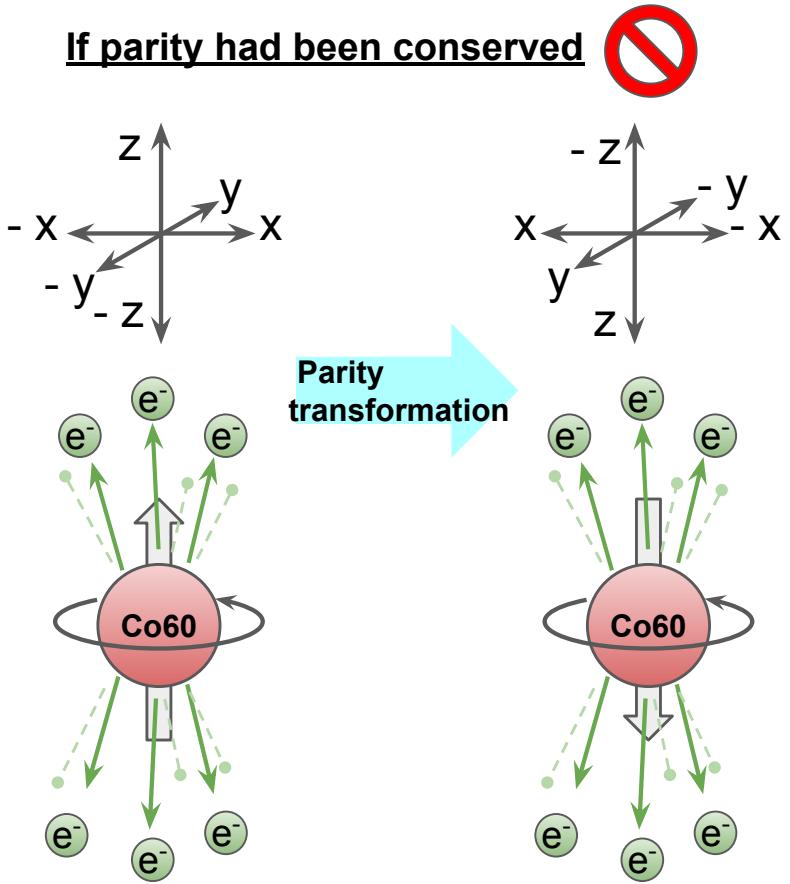


2. What is Parity Symmetry?

Because nuclear spin is an angular momentum, its sign does not flip under parity transformation.

If parity were conserved in the weak force, electrons in the Wu experiment should have been emitted with equal frequency in both directions relative to the nuclear spin.

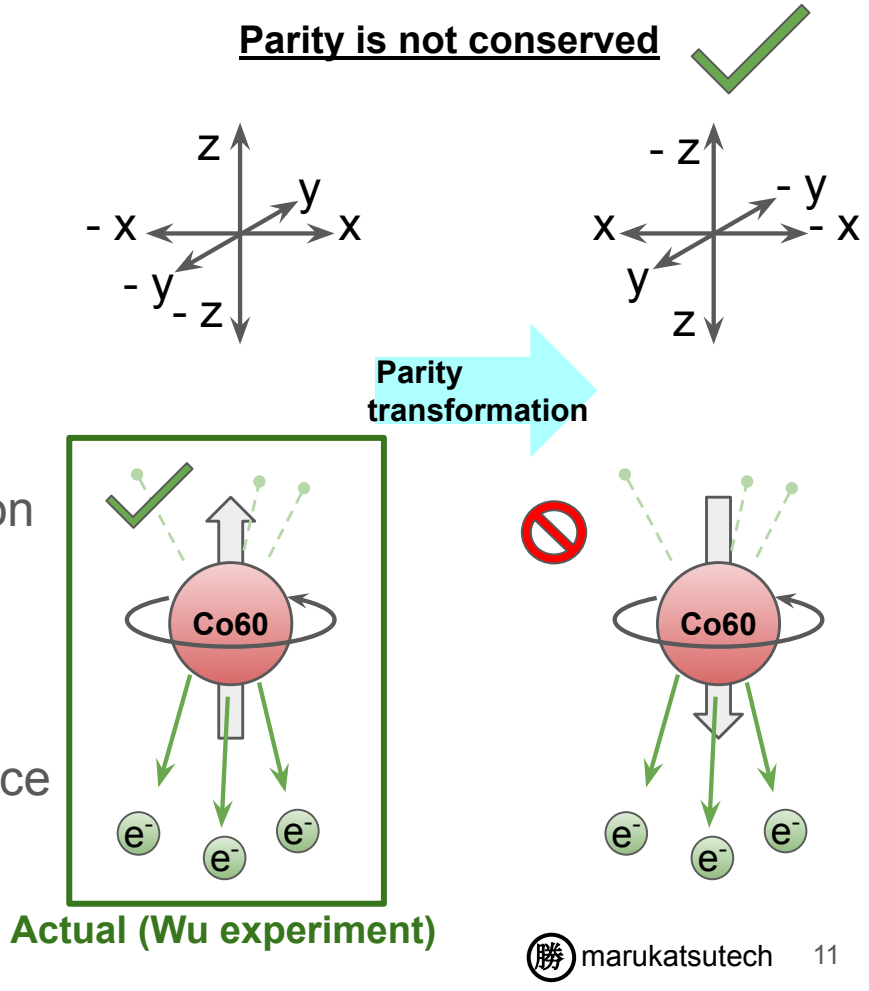
Equal emission in both directions would mean the physical phenomenon remains symmetric even if the signs and arrow directions of momentum relative to nuclear spin reverse.



2. What is Parity Symmetry?

However, the Wu experiment showed that electrons are emitted preferentially in one direction, proving that parity symmetry is broken in β-decay governed by the weak force.

Subsequent experiments (angular correlation and polarization in β-decay, and the 1958 Goldhaber–Grodzins–Sunyar experiment measuring neutrino helicity) and theoretical developments established that the weak force couples exclusively to the left-chiral component of particles.

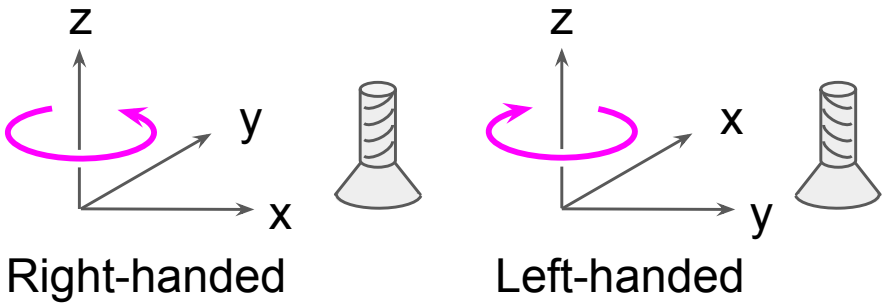


3. What is Chirality?

Chirality is the property whereby an object cannot be superimposed on its mirror image, analogous to the relationship between a right hand and a left hand.

Chirality in quantum mechanics similarly represents particle left-right symmetry, categorized into right-chiral and left-chiral.

However, it incorporates relativistic temporal symmetry in addition to spatial symmetry.



Lorentz symmetry

Chirality in Quantum mechanics

3. What is Chirality?

Quantum mechanics initially lacked special relativity.

Paul Dirac introduced special relativity into quantum mechanics to establish relativistic quantum mechanics, publishing the Dirac equation in 1928.

In non-relativistic quantum mechanics, wave functions or quantum states of spin-1/2 particles (such as electrons and quarks) are represented by 2-component spinors (2-complex-number vectors).

Schrödinger equation

2-component spinor

$$i\hbar \frac{\partial \psi(x,t)}{\partial t} = \hat{H}\psi(x,t) \quad \psi(x) = \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \end{pmatrix}$$

3. What is Chirality?

In the Dirac equation, they are represented by 4-component spinors called Dirac spinors.

Chirality is a relativistic quantum mechanical property describing whether a Dirac spinor is "right-handed" or "left-handed".

relativistic energy–momentum relation

$H^2 = p^2 + m^2$

Dirac equation

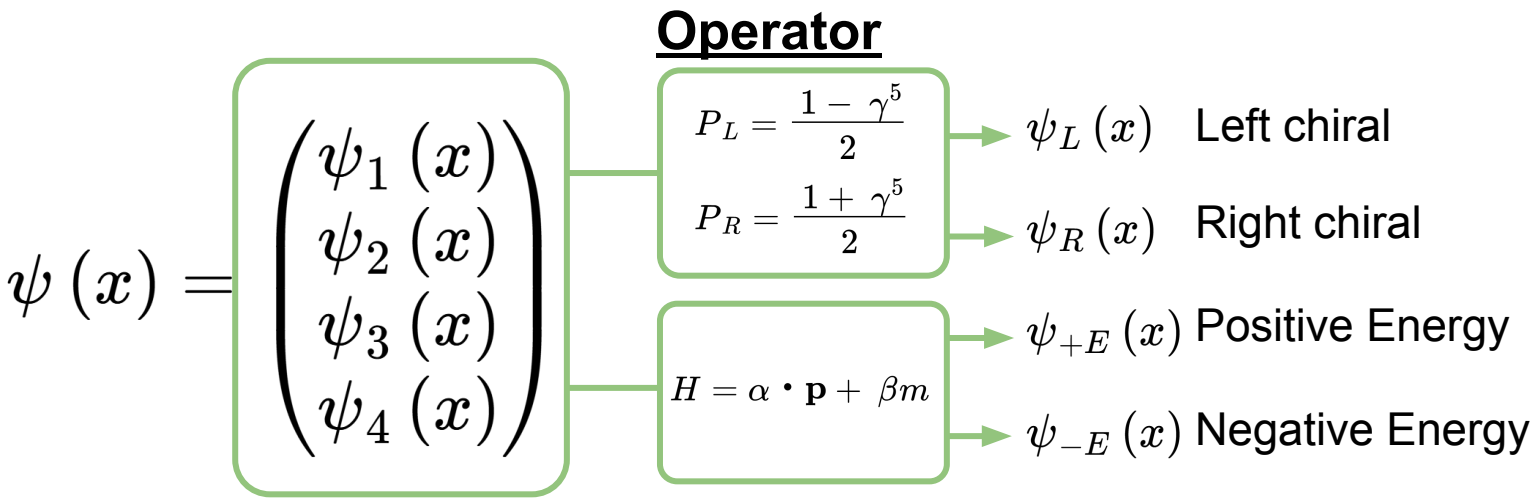
$$(i\gamma^\mu \partial_\mu - m)\,\psi(x) = 0$$

4-component spinor

$$\psi(x) = \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \\ \psi_3(x) \\ \psi_4(x) \end{pmatrix}$$

3. What is Chirality?

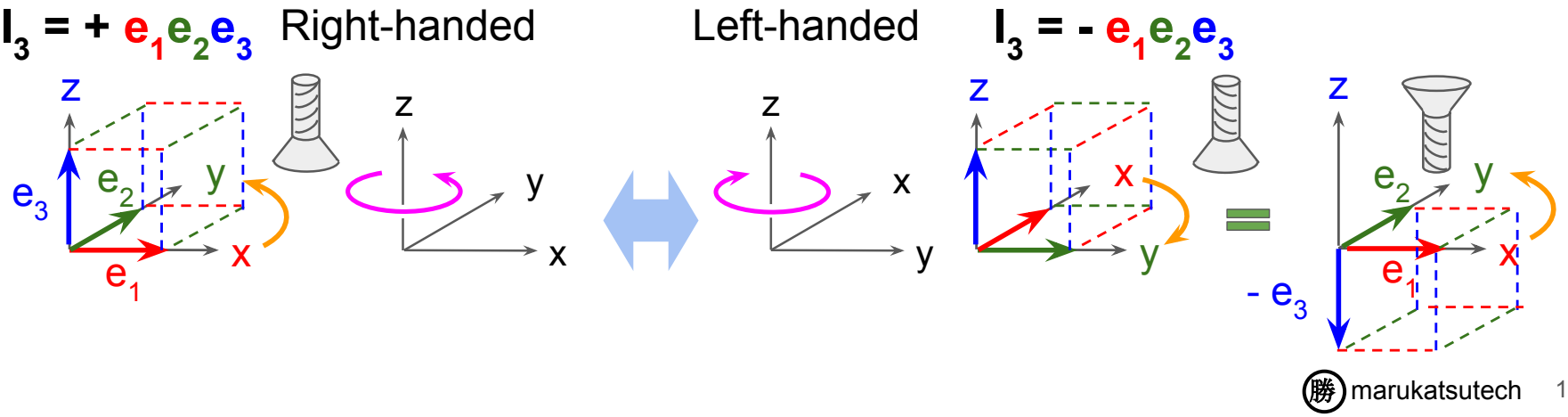
A Dirac spinor contains left-chiral, right-chiral, particle (positive energy), and anti-particle (negative energy) components; for massive particles, these components exist as a mixed state that can be decomposed using projection operators.



3. What is Chirality?

The gamma matrices ($\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$) appearing in the Dirac equation and projection operators (P_L, P_R) represent spacetime orientation.

As a 3D spatial representation of orientation, there is an oriented volume element that attaches orientation (a sign) to the 3D volume element ($I_3 = e_1\, e_2\, e_3$).



3. What is Chirality?

The oriented volume element relates directly to right-handed and left-handed systems: “+$e_1e_2e_3$” for right-handed and “-$e_1e_2e_3$” for left-handed.

While “e_1, e_2, e_3” are orthogonal basis unit vectors, “$\gamma^{0,1,2,3}$” are 4 × 4 matrices containing Pauli matrices (σ).

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

$$\gamma^0 = \begin{pmatrix} \overbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}^{I} & \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} & \underbrace{\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}}_{-I} \end{pmatrix} \quad \gamma^1 = \begin{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} & \overbrace{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}^{\sigma^1} \\ \underbrace{\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}}_{-\sigma^1} & \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \end{pmatrix}$$

$$\gamma^2 = \begin{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} & \overbrace{\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}}^{\sigma^2} \\ \underbrace{\begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}}_{-\sigma^2} & \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \end{pmatrix} \quad \gamma^3 = \begin{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} & \overbrace{\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}}^{\sigma^3} \\ \underbrace{\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}}_{-\sigma^3} & \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \end{pmatrix}$$

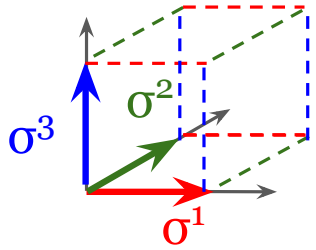
3. What is Chirality?

Pauli matrices serve as generators of rotational operations representing axes of rotation (such as spin orientation), and γ^5 can be viewed as the 4D version (time + 3D spatial dimensions) of an oriented volume element incorporating Lorentz symmetry

If the weak force interacts only with left-chiral components of particles, why are electrons emitted opposite to nuclear spin?

The answer lies in the relationship between helicity and chirality.

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \simeq$$



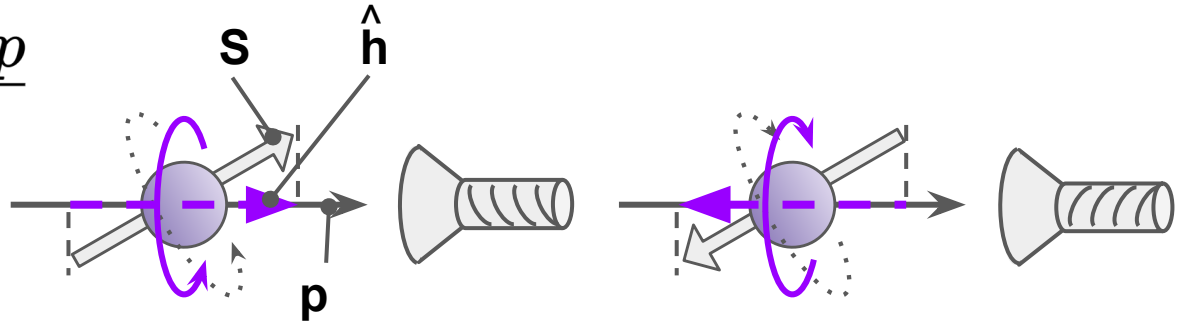
+ Lorentz symmetry

4. Relationship Between Helicity and Chirality

Helicity is a physical quantity indicating whether a particle's spin orientation is aligned with or opposite to its direction of motion.

Mathematically, it is the projection of spin angular momentum onto the momentum vector; it is positive (right-handed) if spin points along momentum, and negative (left-handed) if spin points opposite to momentum.

$$\hat{h} = \frac{S \cdot p}{|p|}$$



h = + (Right handed)

h = - (Left handed)

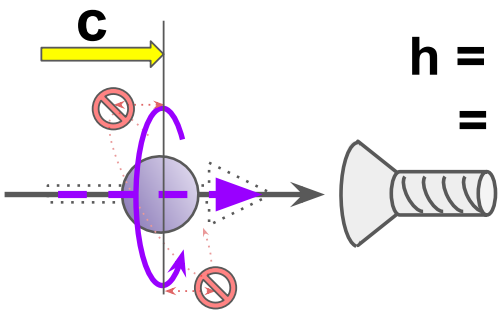
4. Relationship Between Helicity and Chirality

Helicity and chirality are distinct concepts, yet closely connected. For massless particles like photons, helicity = chirality. Chirality is Lorentz-invariant (unchanged by relative observer velocity).

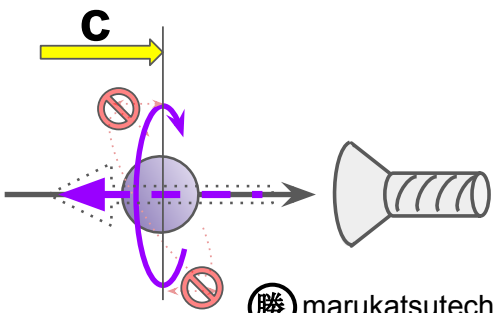
Because massless particles travel at light speed “c”, no observer can overtake them (they have no rest frame).

This restriction on degrees of freedom along the line of motion dictates that helicity equals chirality.

h = + (Right handed)
= right chiral



h = - (Left handed)
= left chiral



4. Relationship Between Helicity and Chirality

Photon helicity corresponds to circular polarization.

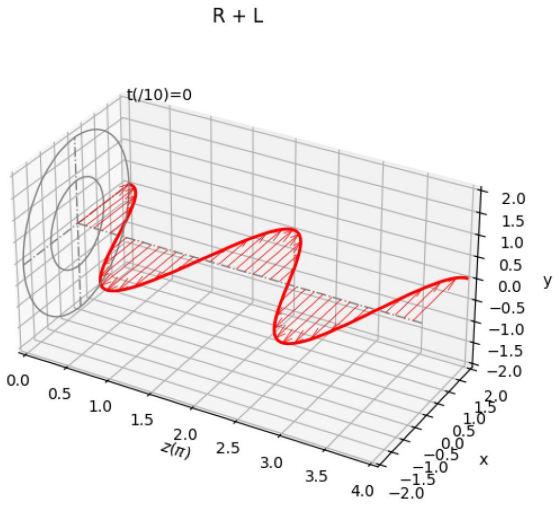
Linear polarization is described as a quantum superposition of equal proportions of helicity “+1” and “-1” photons.

Right and left definitions are reversed by convention between quantum mechanics and optics/electromagnetism.

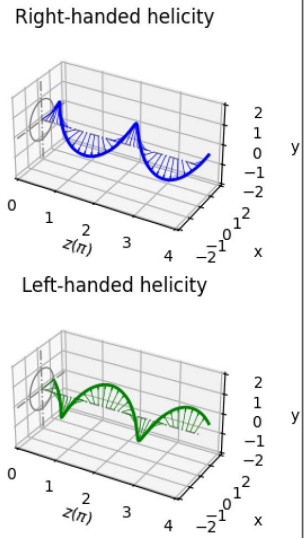
In quantum mechanics, rotation advancing a right-hand screw is right-handed (positive), while rotation advancing a left-hand screw is left-handed (negative).

In optics/electromagnetism, rotation is defined looking from the perspective of light entering the observer's eye.

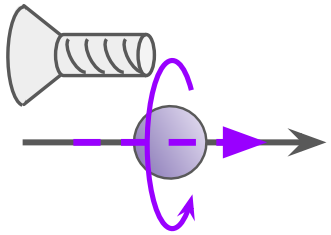
4. Relationship Between Helicity and Chirality



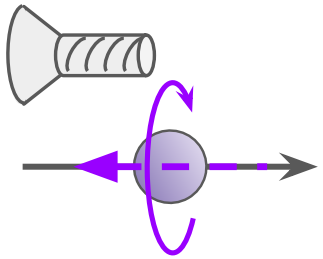
Optics and electromagnetism



Quantum mechanics



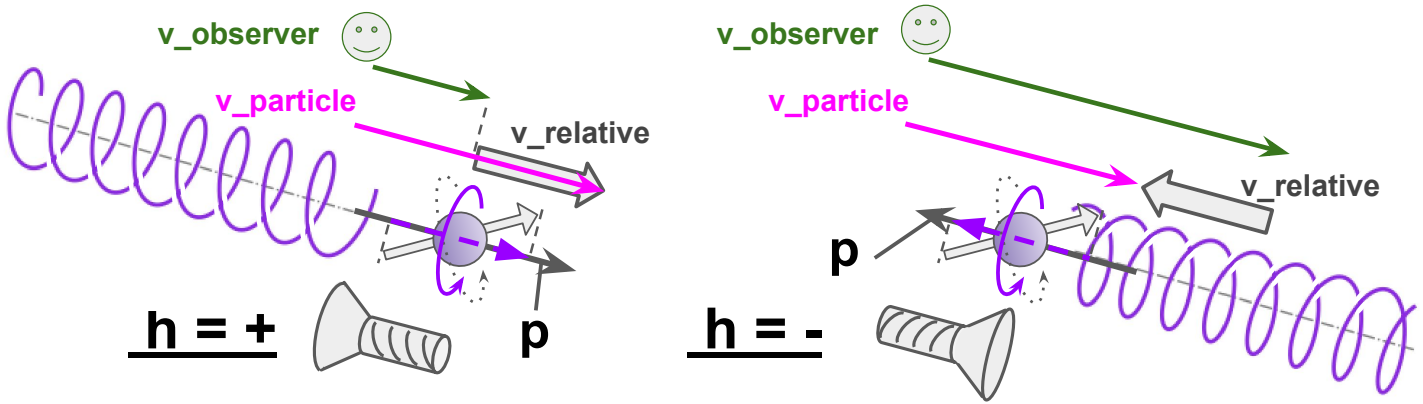
h = + (Right handed)
= right chiral



h = - (Left handed)
= left chiral

4. Relationship Between Helicity and Chirality

For massive particles, helicity ≠ chirality.Massive particles move below light speed, meaning apparent spin orientation changes depending on the observer's relative velocity.



Note: strictly speaking, the spin orientation also changes under a Lorentz transformation due to the Wigner rotation.

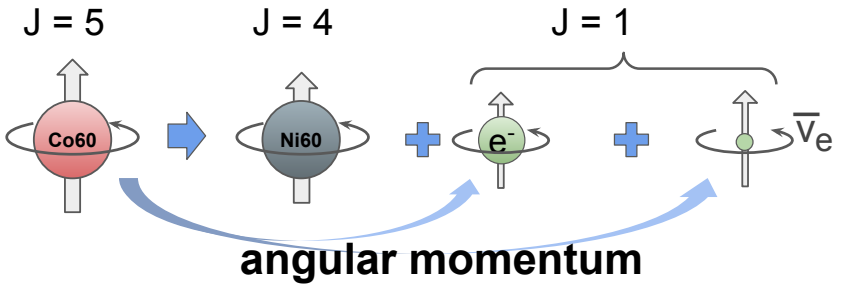
However, for massive particles moving near light speed, helicity ≃ chirality, as high velocity (momentum) reduces mass effects.

4. Relationship Between Helicity and Chirality

In the Wu experiment, Cobalt-60 transitions to Nickel-60 via β-decay ($^{60}Co \rightarrow {}^{60}Ni^{*} + e^{-} + \text{anti-}\nu_e$), dropping from nuclear spin J = 5 to J = 4.

This angular momentum difference “J = 1 (= 5 - 4)” is shared between the electron and electron antineutrino.

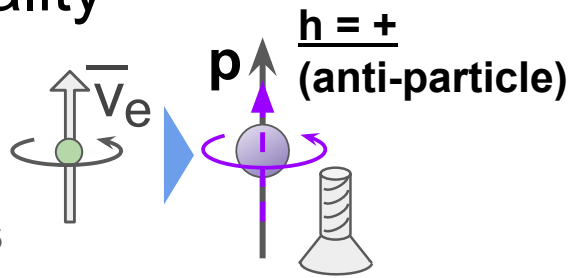
$^{60}Co \rightarrow {}^{60}Ni^{*} + e^{-} + \text{anti-}\nu_e$



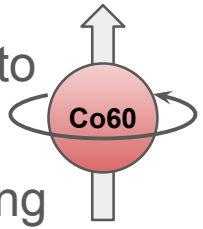
Both particles are emitted near light speed, making helicity ≃ chirality. Because the weak force acts only on left-chiral components, the emitted electron exhibits left helicity (≃ left-chiral component).

4. Relationship Between Helicity and Chirality

On average, spins of both electron and electron antineutrino align with nuclear spin. Left helicity is negative (momentum opposite to spin), causing electrons to be emitted opposite to nuclear spin.

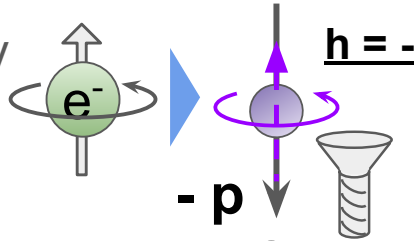


For electron antineutrinos, the weak force also couples to left-chiral components, but being anti-particles, left-right orientation reverses, producing right helicity emitted along nuclear spin.



The weak interaction couples only to the left-chiral component.
≃ Left handed helicity (h = -)

This constitutes the Standard Model explanation of parity violation. However, as noted, why the weak force acts only on left-chiral components remains an unexplained experimental.

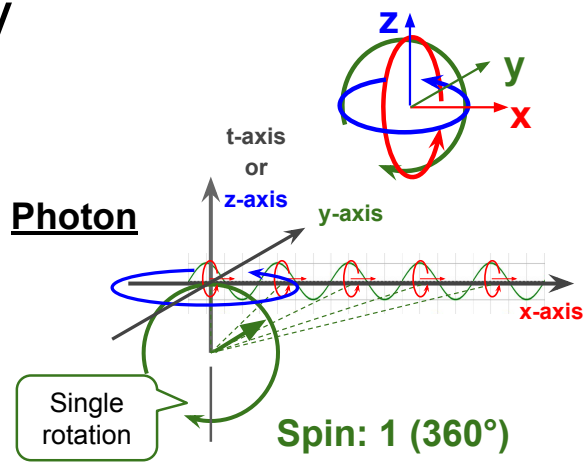


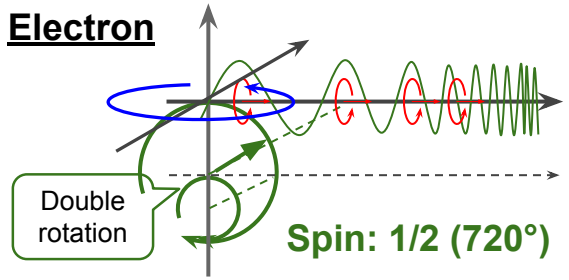
5. Helicity and Chirality in R-space Theory

First consider photons. In R-space Theory, particles are fundamentally composed of three rotation vectors (angular momentum) “x, y, z” corresponding to 3 dimensions.

These rotation vectors establish mutual phase relations; for instance, “vector-x” projects into space via phases of vectors “y” and “z”.

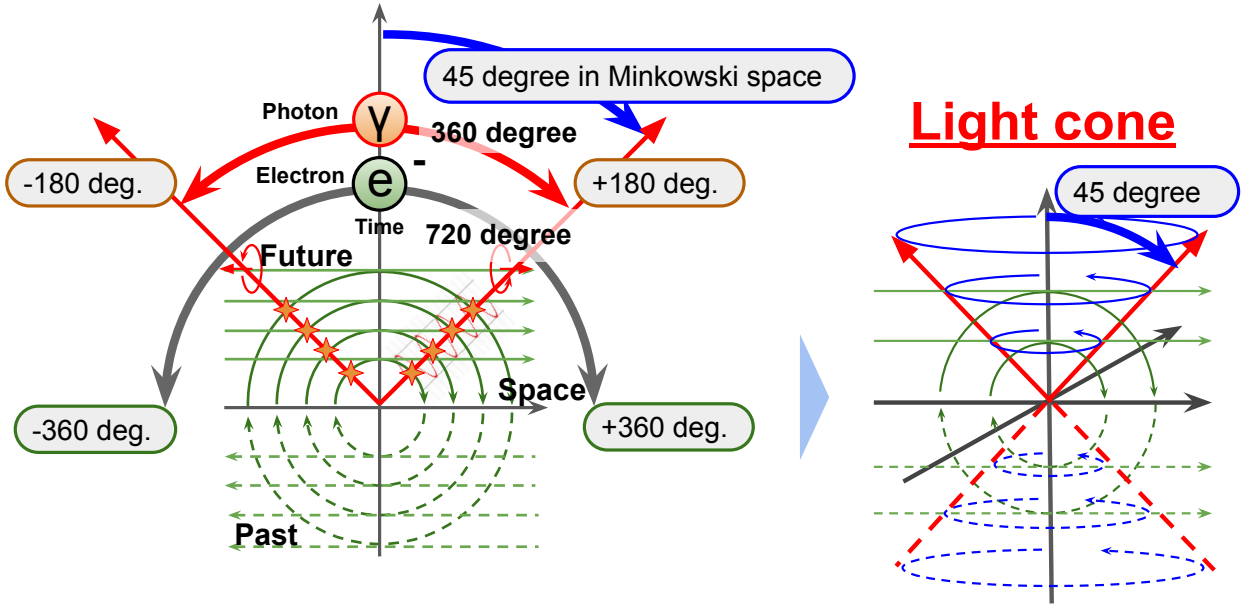
Photons exhibit single-rotation symmetry (1 rotation per 360°), while electrons and quarks exhibit double-rotation symmetry, i.e., 1/2-rotation symmetry (1 rotation per 720°).





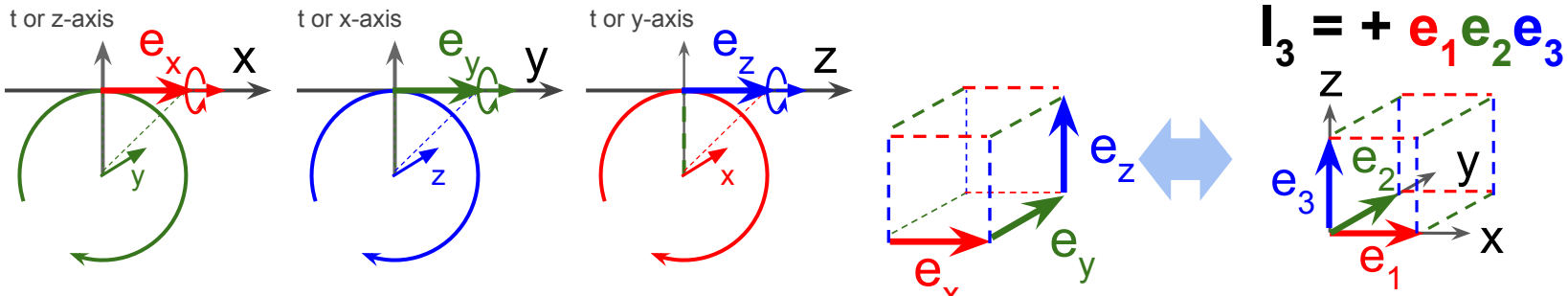
5. Helicity and Chirality in R-space Theory

Due to symmetry constraints, interactive phases are restricted, causing photons to interact exclusively on the light cone.

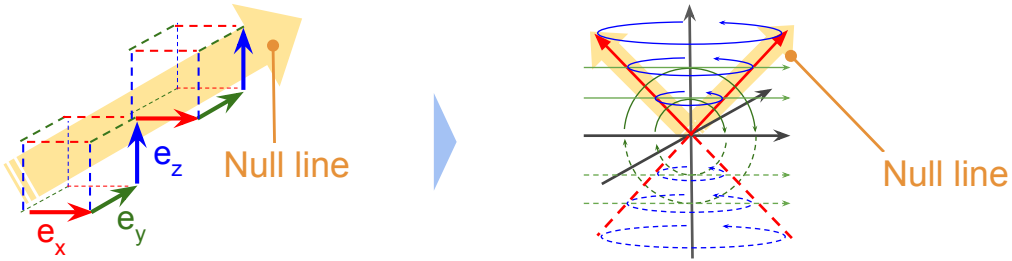


5. Helicity and Chirality in R-space Theory

Applying oriented volume elements to the R-space photon model, unit phase-projection vectors (e_x, e_y, e_z) correspond to "e_1, e_2, e_3".



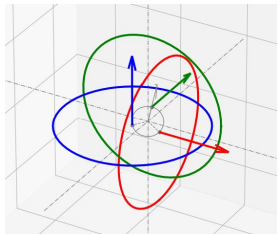
Photons propagate cyclically as "e_x" to "e_y" to "e_z" to "e_x" to "e_y" to "e_z" ……. Arranging this along a specific axis (e.g., x-axis) forms the photon worldline (null-line).

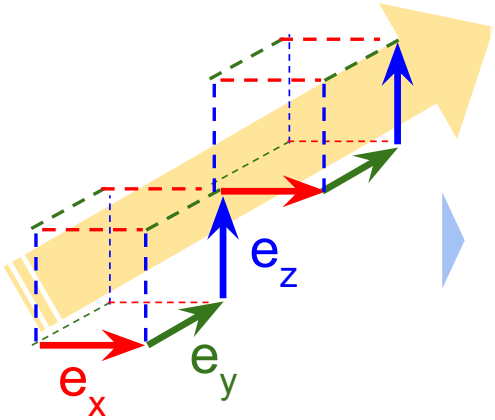


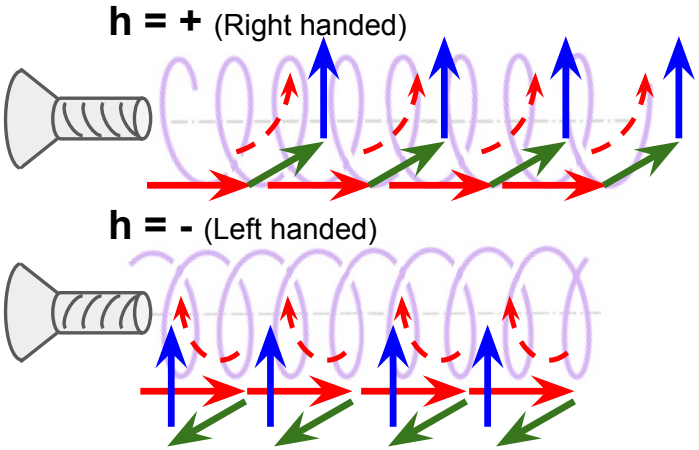
5. Helicity and Chirality in R-space Theory

“e_y” and “e_z” generate phase rotation around the x-axis, forming a helix that defines photon helicity.

If “$e_x e_y e_z$” represents a right-handed system (h = +1), reversing “e_y” gives “$e_x(-e_y)e_z = -e_x e_y e_z$”, producing a left-handed system (h = -1).

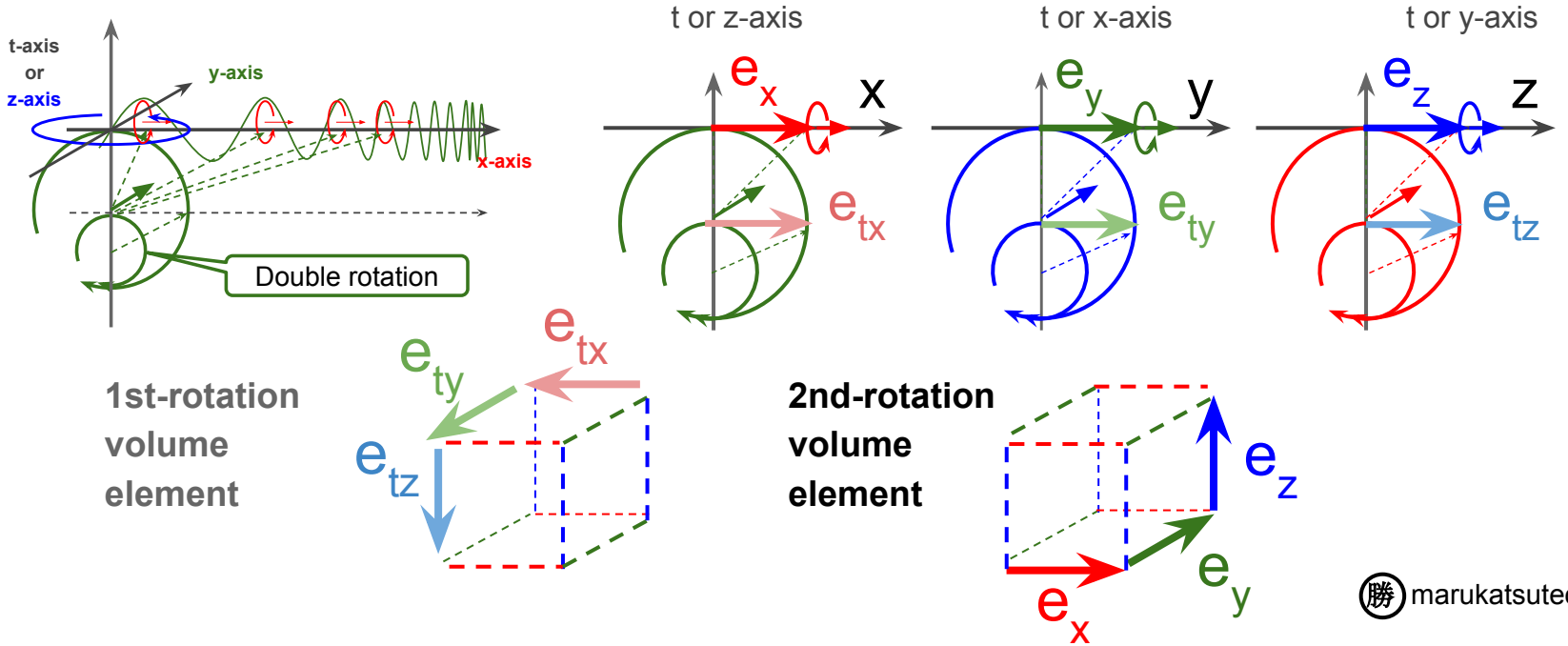






5. Helicity and Chirality in R-space Theory

Applying oriented volume elements to massive particles yields two volume elements due to double rotation: the first-rotation volume element "$e_{tx}e_{ty}e_{tz}$" and the second-rotation volume element "$e_xe_ye_z$".

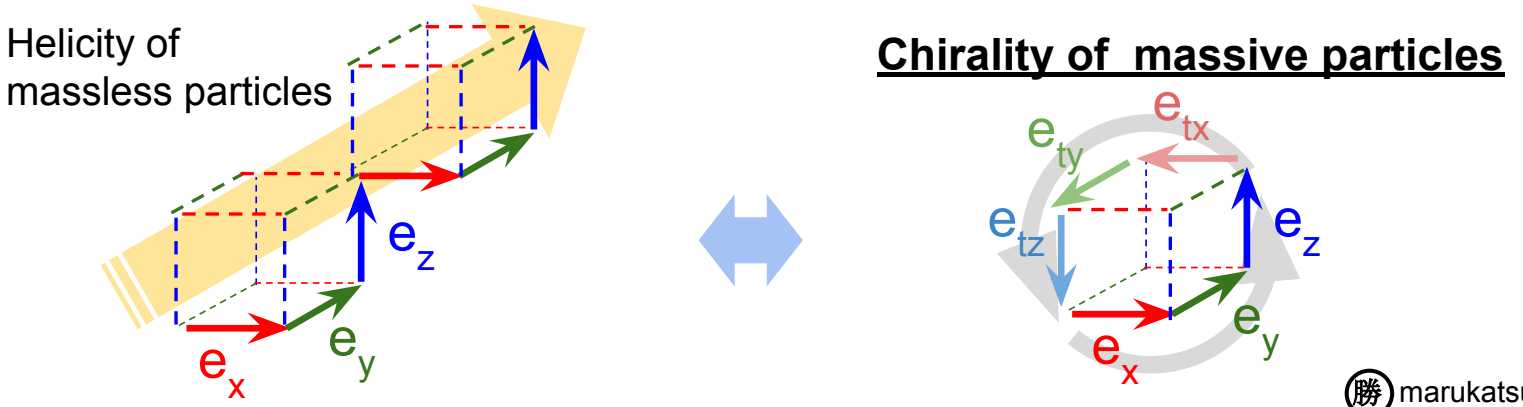


5. Helicity and Chirality in R-space Theory

Double rotation forms a closed circulation connecting the endpoint of “e_z” to the startpoint of “e_{tx}”, and the endpoint of “e_{tz}” to the startpoint of “e_x”.

This structure represents chirality in massive particles (a mixed state of right- and left-chiral).

Because start- and endpoints coincide, it circulates without spreading out.

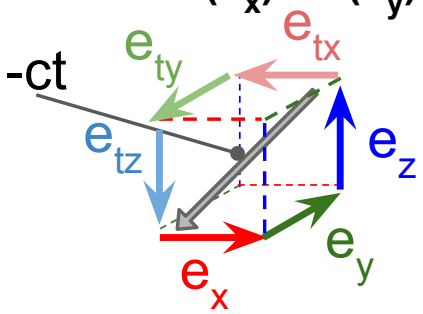


5. Helicity and Chirality in R-space Theory

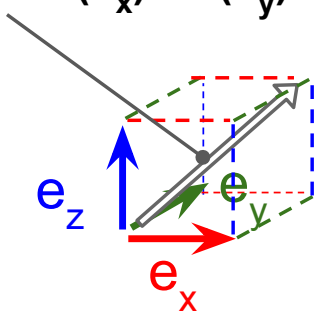
Letting “ct” be the vector connecting the startpoint of “e_{tx}” and endpoint of “e_{tz}” in the first volume element, its length equals the diagonal of the second volume element: “$c^2t^2 = (e_x)^2 + (e_y)^2 + (e_z)^2$”.

Vector directions in the first element oppose the second, appearing as “$-c^2t^2$” from the second element's perspective. Constructing a 4D volume element including “$-c^2t^2$” gives “Volume Element = $-c^2t^2 + (e_x)^2 + (e_y)^2 + (e_z)$”, matching the Minkowski spacetime interval equation “$s^2 = -c^2t^2 + x^2 + y^2 + z^2$”.

$$s^2 = - c^2t^2 + (e_x)^2 + (e_y)^2 + (e_z)^2$$



$$c^2t^2 = (e_x)^2 + (e_y)^2 + (e_z)^2$$

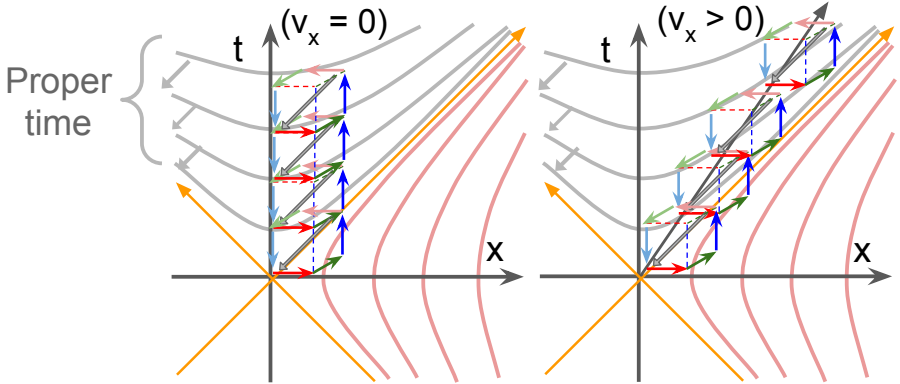


5. Helicity and Chirality in R-space Theory

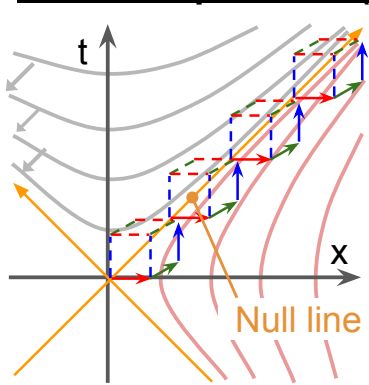
The first-rotation volume element corresponds to proper time of a massive particle. Two volume elements mark proper time while circulating; lines of equal proper time form hyperbolas on Minkowski diagrams.

Lacking a first rotation, photons yield “$0 = -c^2t^2 + (e_x)^2 + (e_y)^2 + (e_z)^2$” to “$0 = -c^2t^2 + x^2 + y^2 + z^2$”, matching the null-line equation of the light cone with zero proper time.

Massive particles



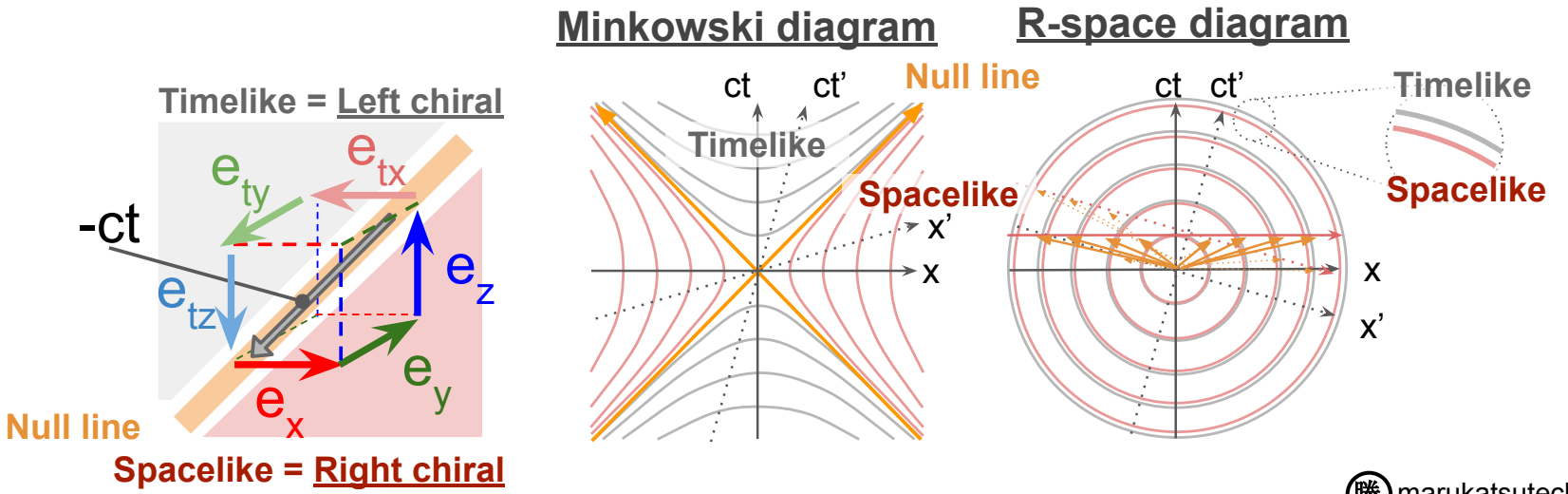
Massless particles (photons)



5. Helicity and Chirality in R-space Theory

Half of this chiral volume element circulation is timelike and half is spacelike, mirroring Timelike/Spacelike regions in Minkowski diagrams.

The phase circle in R-space diagrams and hyperbola in Minkowski diagrams represent the same structure.

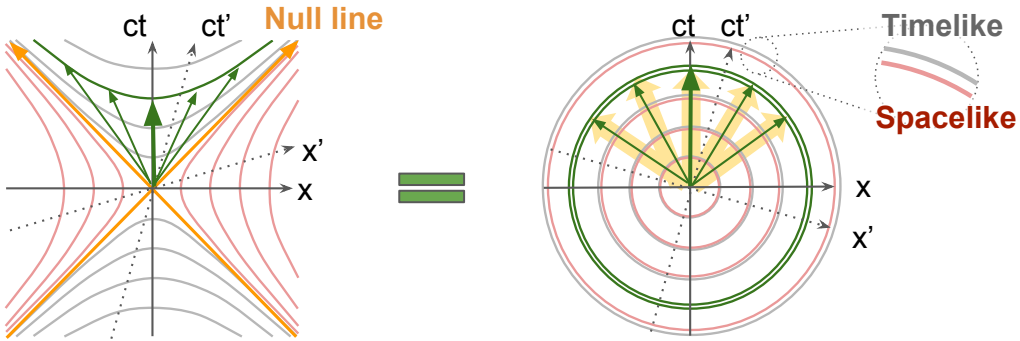


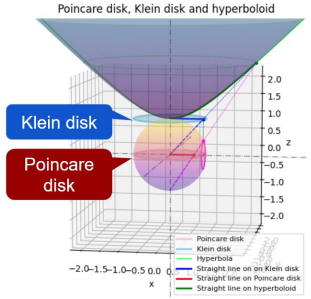
5. Helicity and Chirality in R-space Theory

In R-space diagrams, time and space are identical, differing only by observer perspective.

Representing temporal and spatial circulations on one phase circle gives “Radius^2 = Time^2 + Space^2”.

Minkowski diagrams separate time and space to measure proper time using invariant light speed, yielding “s^2 = -Time^2 + Space^2” (hyperbola).





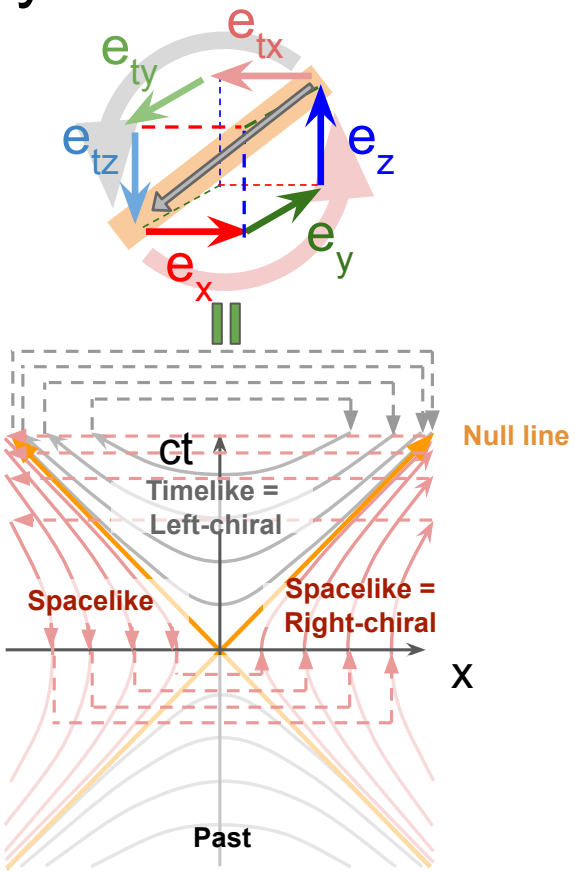
5. Helicity and Chirality in R-space Theory

Spatial circulation corresponds to right-chiral, while temporal circulation corresponds to left-chiral.

The first rotation of double rotation generates time.

Proper-time circulation (first rotation) is left-chiral, circulating within the Timelike region.

The second rotation is right-chiral, circulating within the Spacelike region.

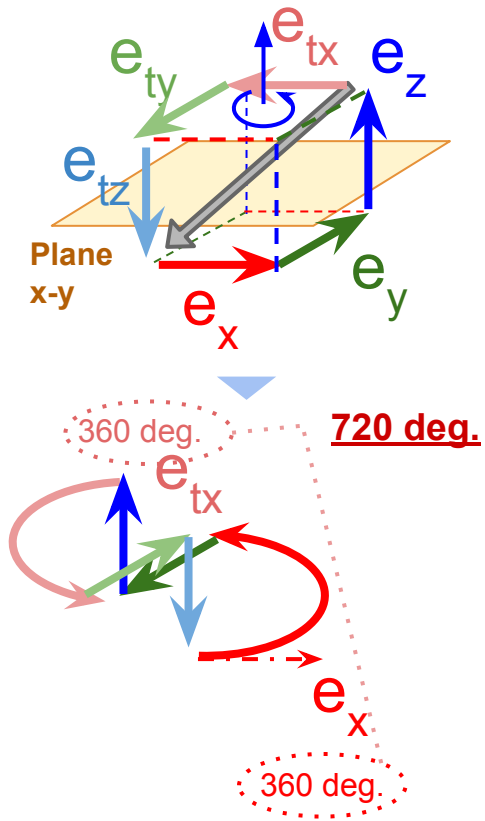


6. Quantum Spin, Chirality, and Helicity in R-space Theory

The relationship between the double Hopf-link electron model and chirality is as follows. Chirality circulates, returning to original spatiotemporal coordinates even when moving through spacetime. Thus, each chiral basis vector circulates individually.

First, x-axis basis vectors “e_{tx}” and “e_x” are bent into semicircles on the x-y plane and connected, preserving orthogonality at connections.

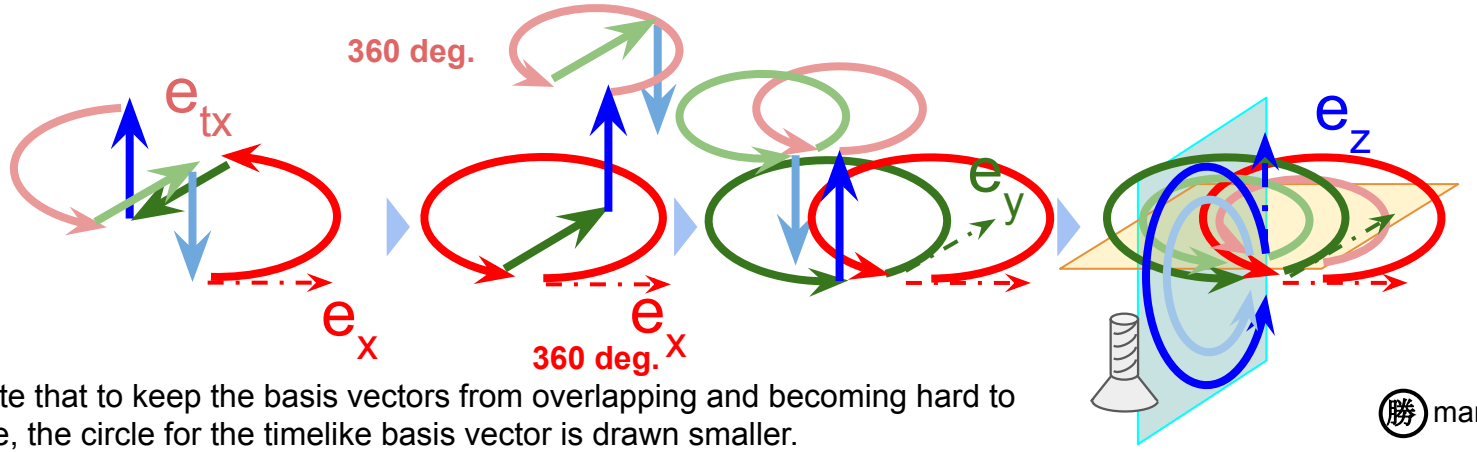
Because temporal and spatial dimensions circulate, each semicircle corresponds to 360°. This spatiotemporal circle equals 720° per revolution, representing spin-1/2 (1/2-rotation symmetry)



6. Quantum Spin, Chirality, and Helicity in R-space Theory

Inside the perceived light cone (Timelike region), spatial circulation is 360°, requiring further bending into a circle. (This temporarily disrupts chiral circulation, which reconnects in the final step).

Next, the same operation is performed on y-axis basis vectors on the x-y plane. Finally, z-axis basis vectors are bent into circles on the y-z plane, reconnecting chiral circulation.

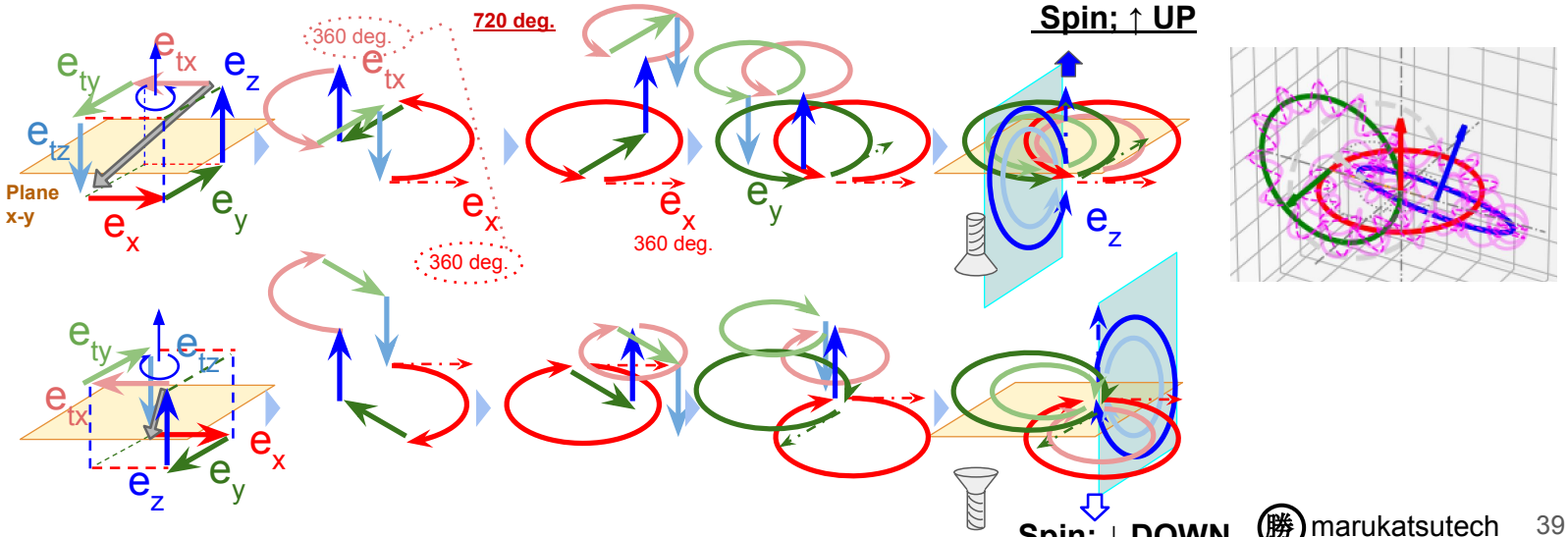


Note that to keep the basis vectors from overlapping and becoming hard to see, the circle for the timelike basis vector is drawn smaller.

6. Quantum Spin, Chirality, and Helicity in R-space Theory

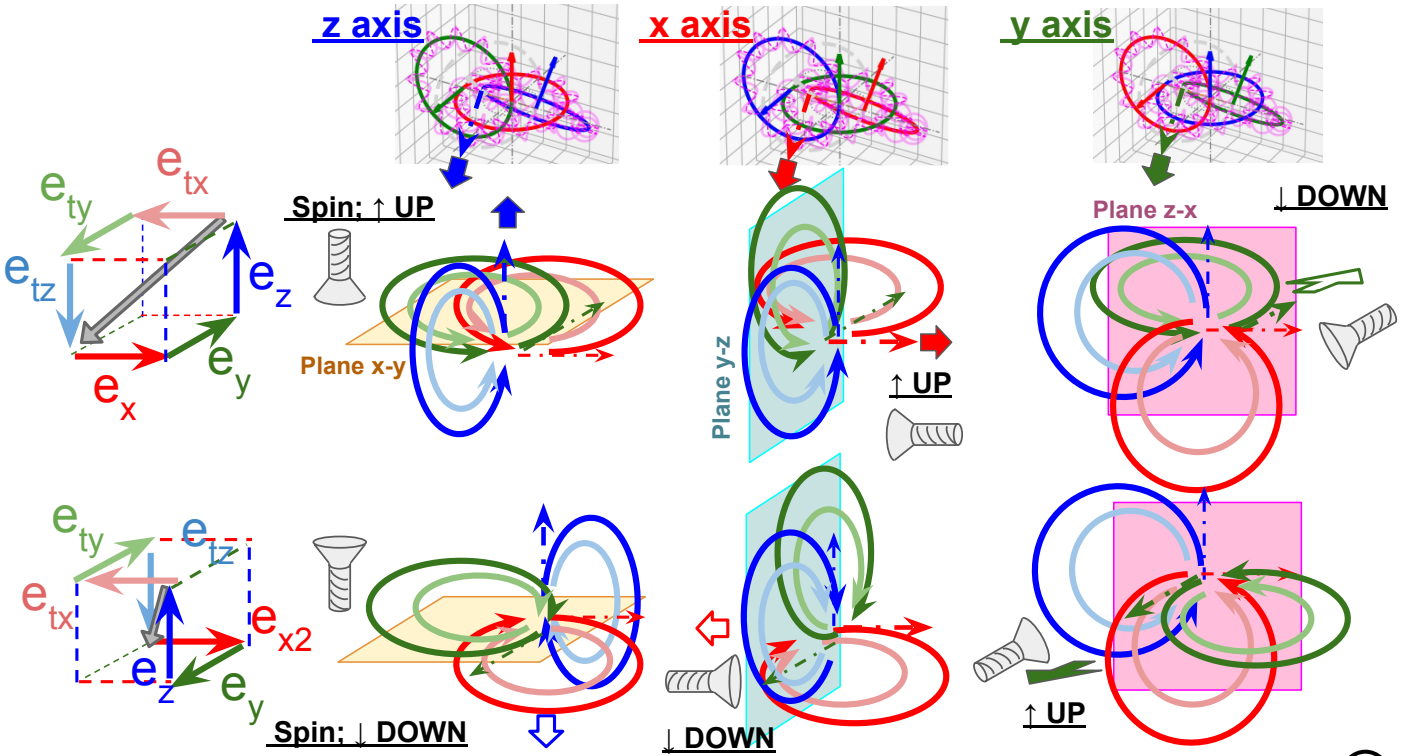
Incorporating precession yields the electron model. Thus, the double Hopf-link electron model represents chiral circulation.

Spin direction: right-handed rotation of "e_x, e_y" relative to "e_z" is Up; left-handed rotation is Down.



6. Quantum Spin, Chirality, and Helicity in R-space Theory

The spin axis is determined by which two axis basis vectors circulate first.



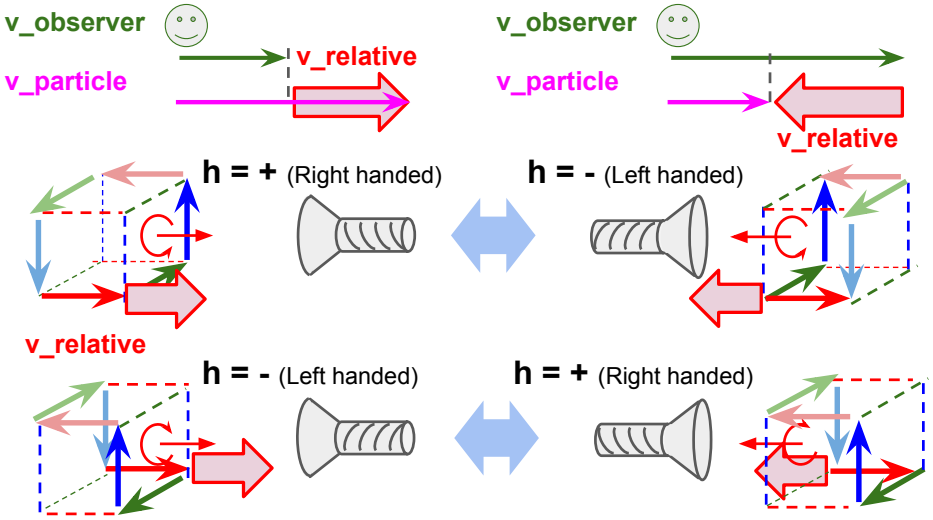
Note that e_ are basis vectors, and any other direction is expressed as a combination of these basis vectors.

6. Quantum Spin, Chirality, and Helicity in R-space Theory

Helicity of massive particles differs from photons due to chiral circulation.

Returning to original coordinates creates a rest frame, establishing relative velocity where helicity represents rotation relative to motion direction.

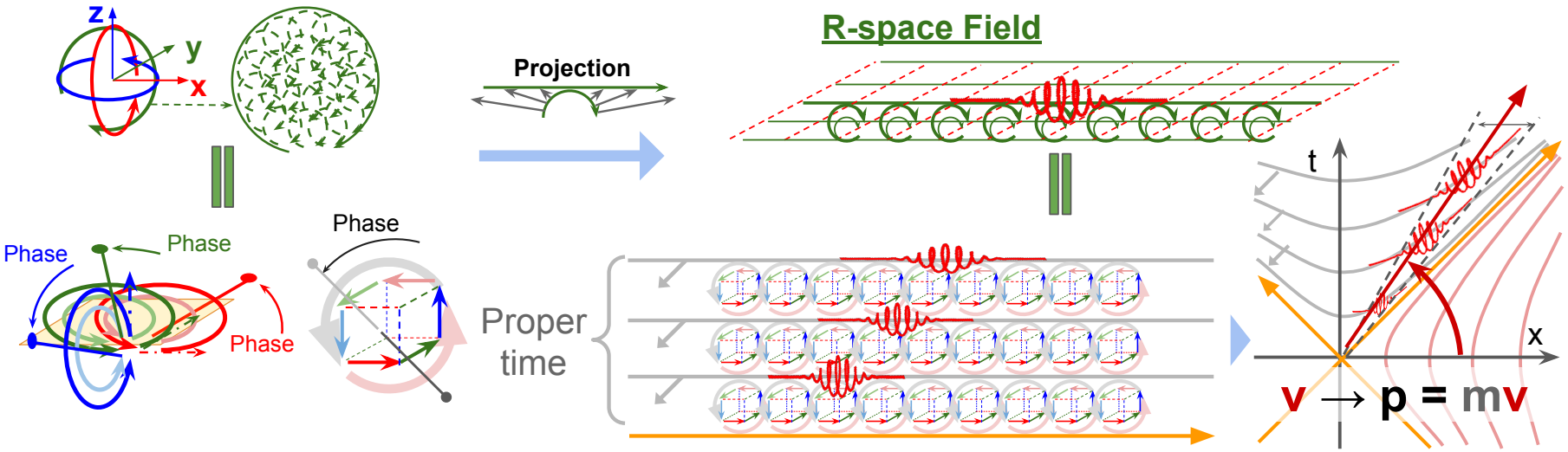
Thus, helicity sign depends on observer relative velocity.



6. Quantum Spin, Chirality, and Helicity in R-space Theory

In R-space Theory, a particle is an angular momentum vortex composed of smaller vortices representing chirality.

Projecting them into 3D space yields the R-space field.

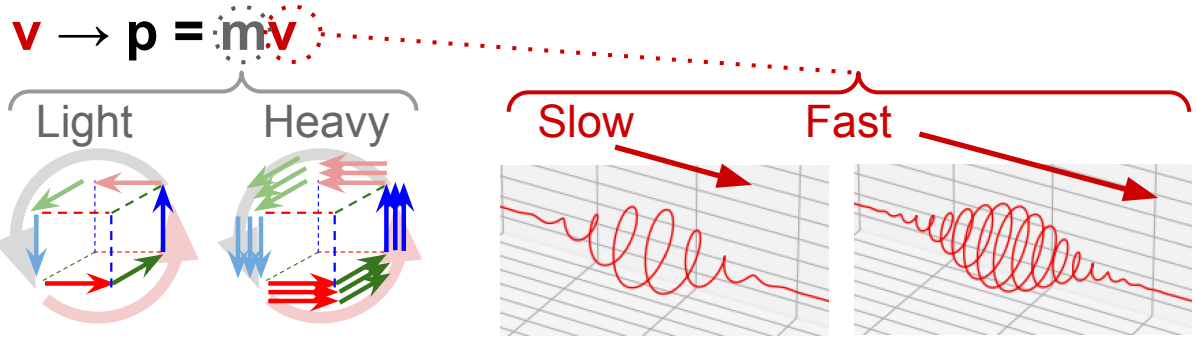


6. Quantum Spin, Chirality, and Helicity in R-space Theory

Chirality phase superposition generates a wave packet whose peak velocity is relative velocity(v); denser spiral spacing means higher velocity.

Rotation frequency (rotation speed) represents mass(m), corresponding to chiral circulation speed (volume element density).

These two densities (spacing density and chirality density) endow the wave packet with momentum “p = mv” and quantum spin with angular momentum.

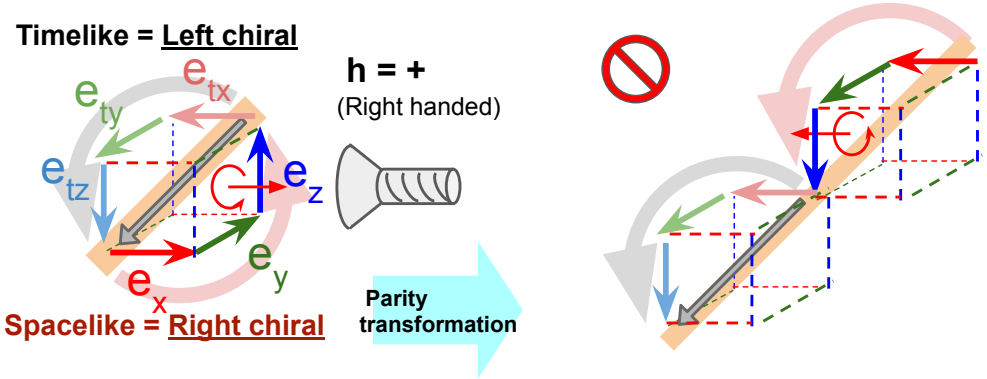


7. Mechanism of Parity Violation in R-space Theory

Performing a parity transformation on R-space chirality replaces coordinates (x, y, z) with (-x, -y, -z).

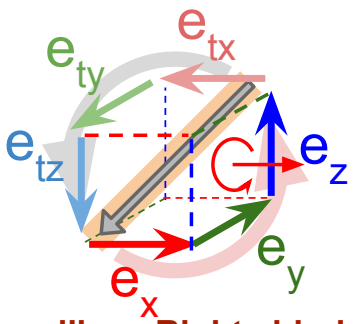
Reversing directions of spacelike basis vectors “e_x, e_y, e_z” breaks chiral circulation. Failure of spacetime circulation breaks spacetime structure itself.

Therefore, weak interactions involving chirality are forced to violate parity symmetry.

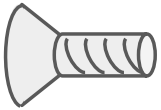


7. Mechanism of Parity Violation in R-space Theory

Timelike = Left chiral



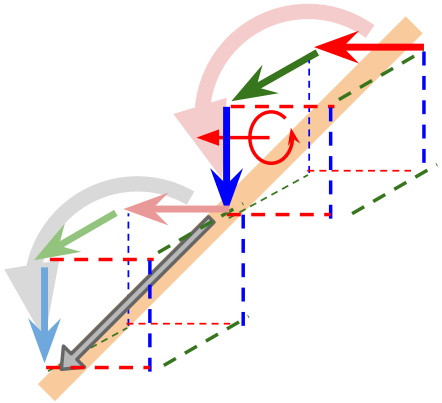
h = +
(Right handed)



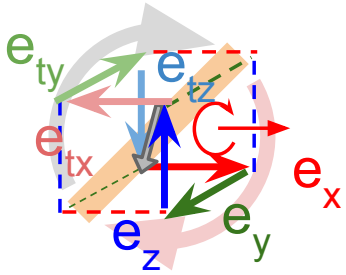
Spacelike = Right chiral



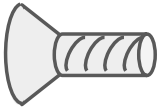
Parity transformation



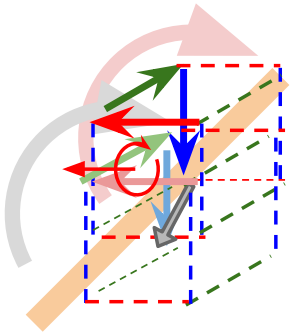
Timelike = Left chiral



h = -
(Left handed)



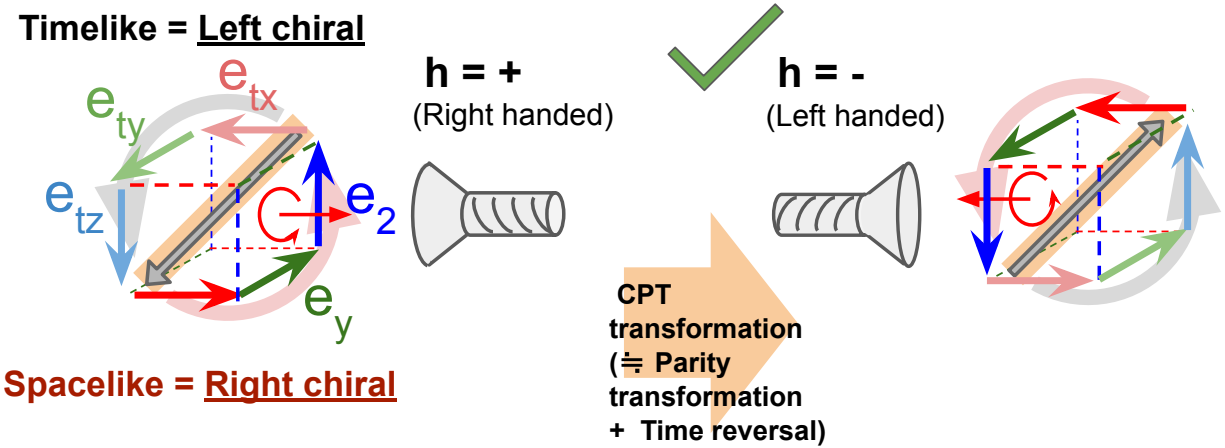
Spacelike = Right chiral



7. Mechanism of Parity Violation in R-space Theory

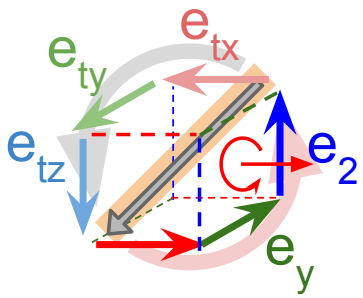
If time is also reversed (reversing timelike basis vectors) alongside parity transformation, spacetime circulation is preserved.

This explains why weak interactions preserve CPT symmetry. (In R-space Theory, rotation vectors represent charge “C”, so reversing spacelike basis vector directions corresponds to “C” transformation).

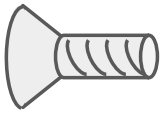


7. Mechanism of Parity Violation in R-space Theory

Timelike = Left chiral



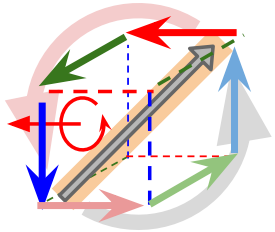
h = +
(Right handed)



Spacelike = Right chiral

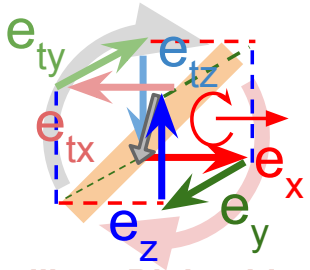
CPT transformation
(≒ Parity transformation + Time reversal)

Spacelike = Right chiral

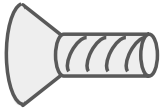


Timelike = Left chiral

Timelike = Left chiral

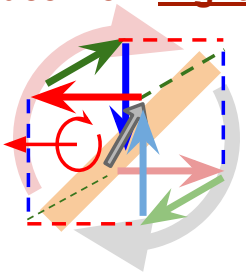


h = -
(Left handed)



Spacelike = Right chiral

Spacelike = Right chiral



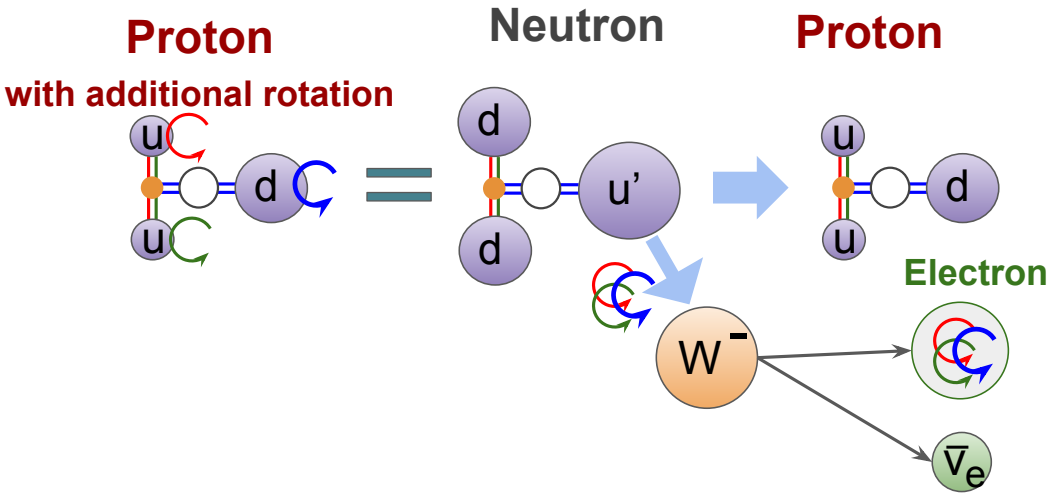
Timelike = Left chiral

8. Explanation of Wu Experiment Results in R-space Theory

Why does the weak force act exclusively on left-chiral components?

In R-space β-decay, when “d,d,u-like” quarks forming a neutron transition to proton “u,u,d”, rotational energy is emitted as a W^- boson.

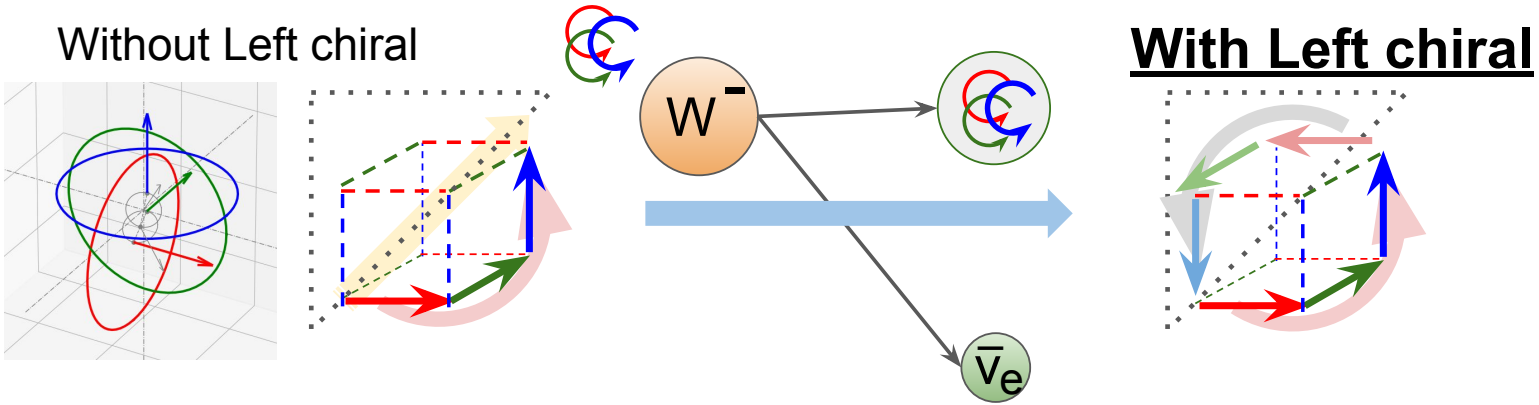
Note: in the Standard Model, one of the u quarks in the neutron “d, d, u” is said to convert into a d quark, turning it into a proton “u, u, d”.



8. Explanation of Wu Experiment Results in R-space Theory

Three rotation vectors of the W^- boson lack chiral circulation (double rotation) and resemble high-energy photons.

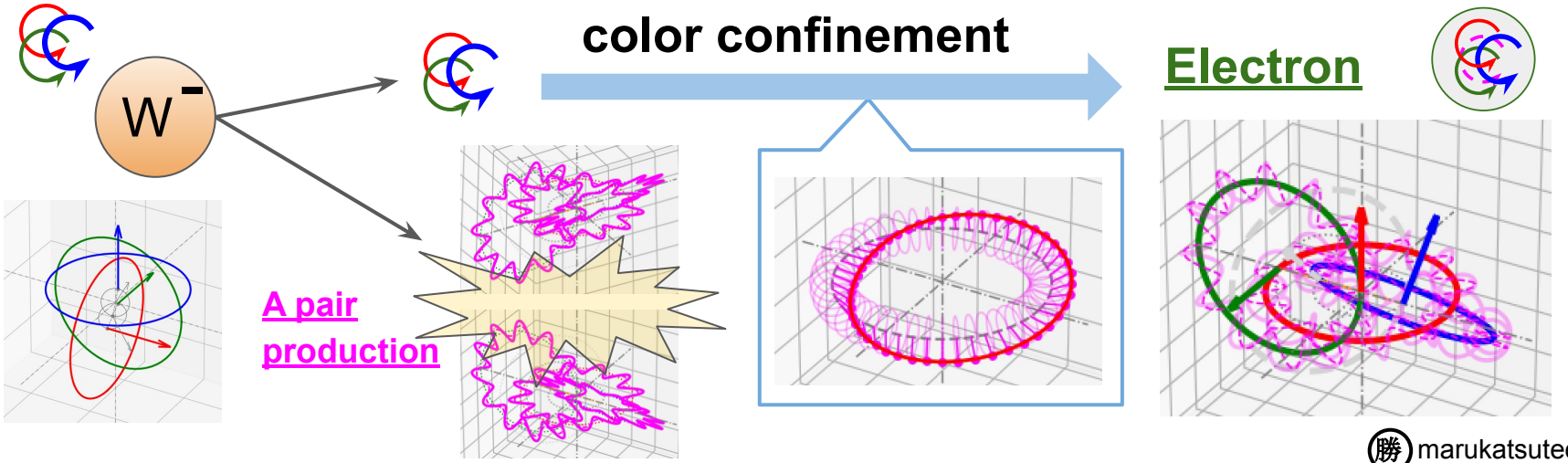
To convert into a stable electron, the W^- boson must acquire a left-chiral component.



8. Explanation of Wu Experiment Results in R-space Theory

Rotation vectors emitted as a W^- boson use energy to pair-produce color charges.

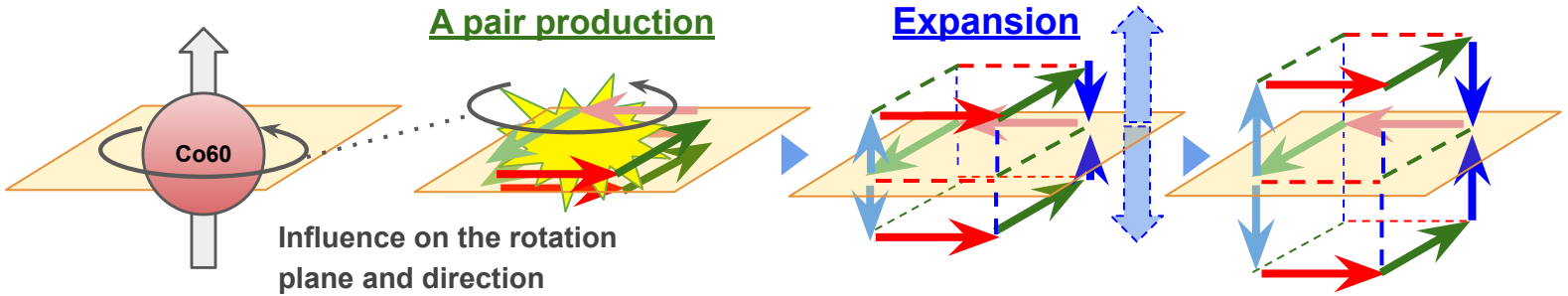
Color charges organize the rotation vectors into a double Hopf link (color confinement), acquiring left-chirality to become an electron.



8. Explanation of Wu Experiment Results in R-space Theory

During pair production, two 2D rotations occur in the two timelike pairs "e_{tx}-e_{ty}" form tangent to the rotation plane.

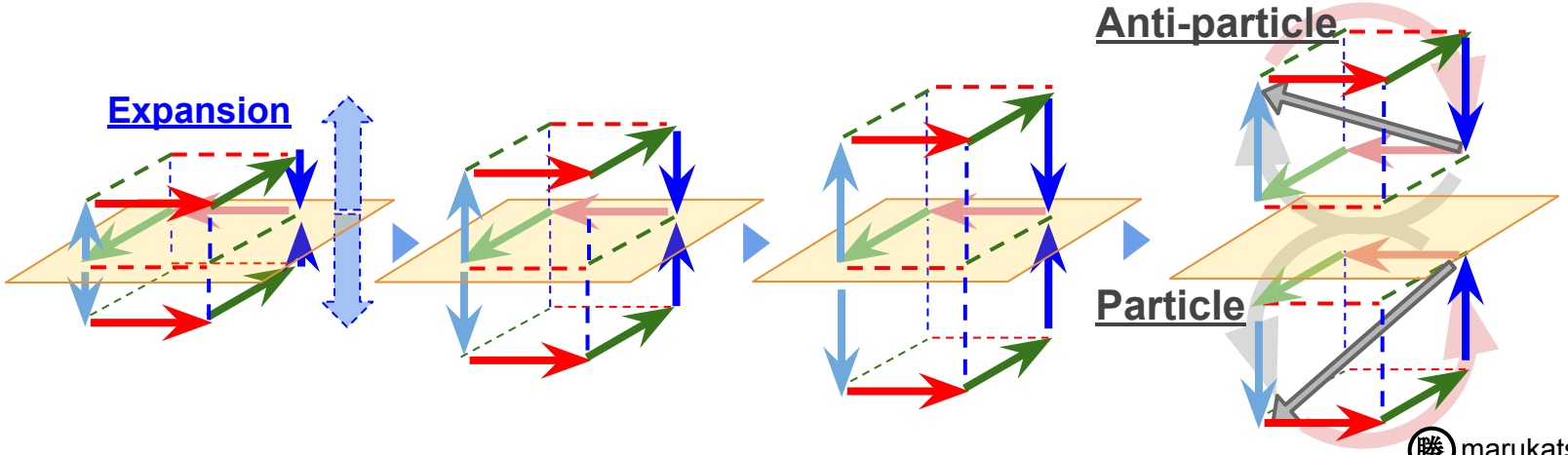
The two spacelike pairs "e_x-e_y" expand away from the rotation plane to separate spatial positions.



8. Explanation of Wu Experiment Results in R-space Theory

When 3D circulation becomes isotropic (equal basis vector lengths), two inverted chiralities complete and separate.

Chirality generated toward the spin direction is the anti-particle; the opposite side is the particle. (These two chiralities are related by an inversion of orientation. They cannot be superimposed by any geometric rotation).

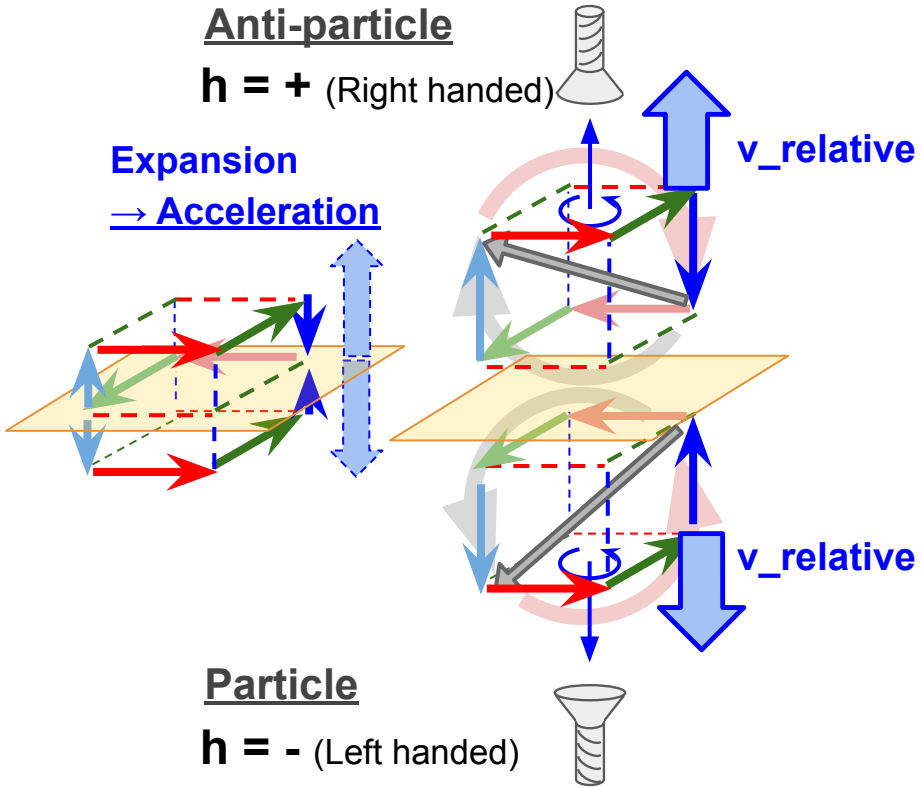


8. Explanation of Wu Experiment Results in R-space Theory

Extension of spacelike vector "e_z" generates acceleration and relative velocity along "e_z".

Consequently, anti-particles are emitted toward nuclear spin (right-handed helicity), and particles opposite to spin (left-handed helicity).

Color charges behave identically.

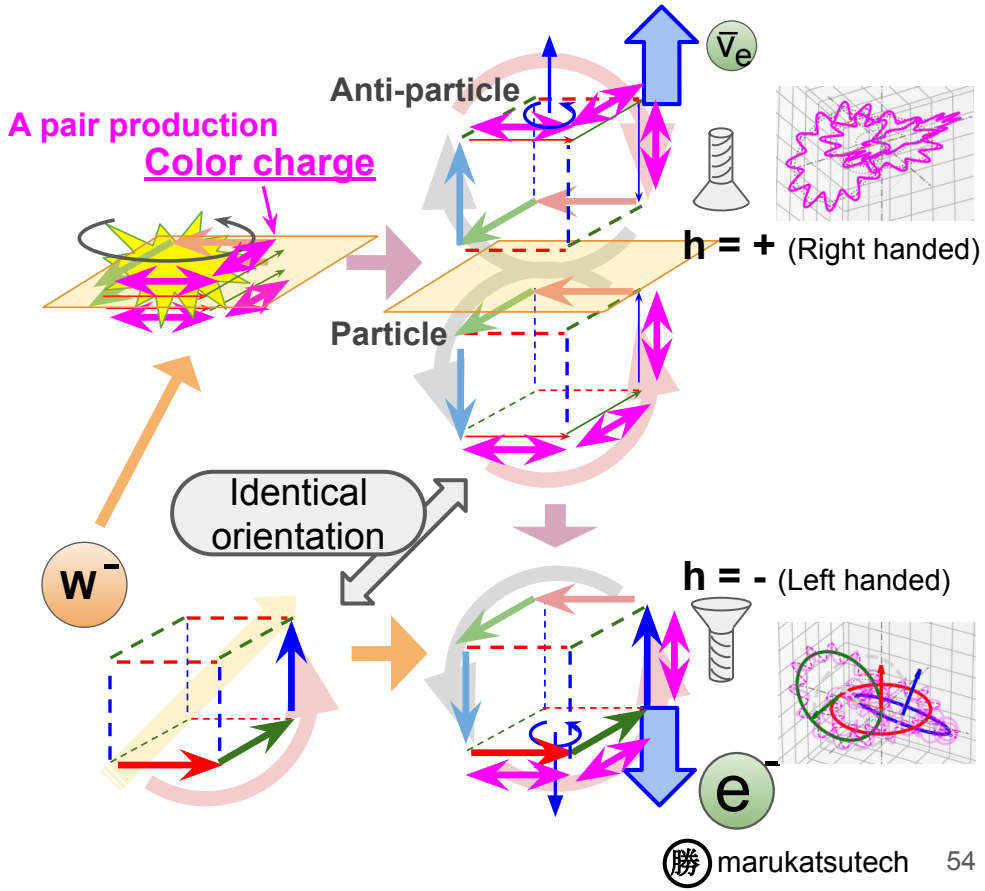


8. Explanation of Wu Experiment Results in R-space Theory

Electrons are particles(not anti-particles), absorbing color charge of identical chiral volume orientation(particles orientation) emitted opposite to nuclear spin.

Absorbing this momentum causes electrons to be emitted opposite to spin.

Anti-particle color charge is emitted along spin as an electron antineutrino.

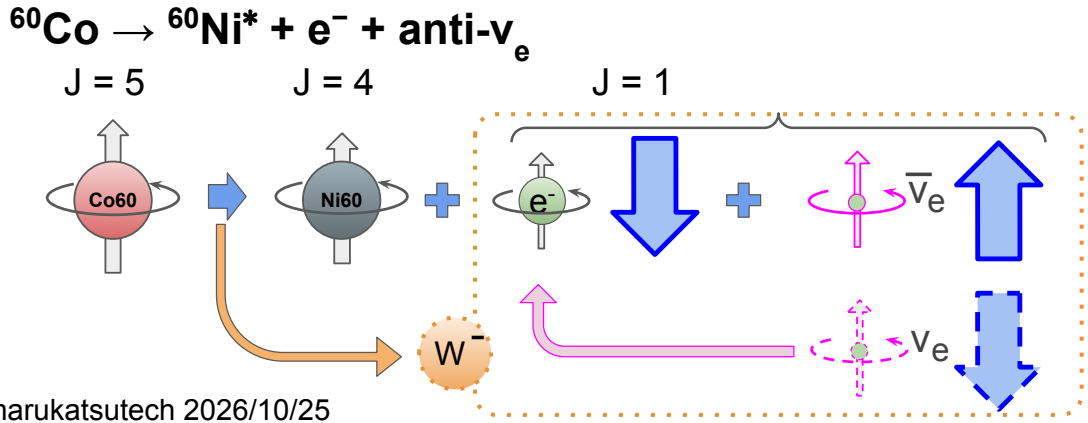


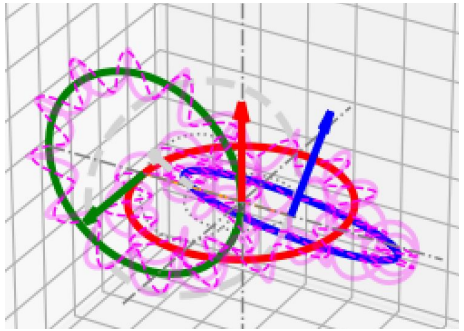
8. Explanation of Wu Experiment Results in R-space Theory

Pair production preserves CPT symmetry, but because the electron absorbs one side, parity symmetry appears broken.

This concludes the explanation of parity violation in the weak force.

Parity violation in the weak force is fundamentally related to electron formation via color confinement.





This is all for my presentation.

Thanks for watching!

